Scientific Computing & Advanced Programming Lab

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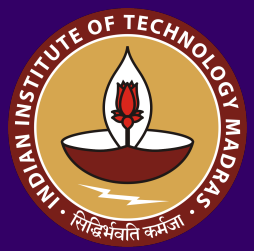
Lecture-1



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Welcome!

- Slides will be available at ”https://sauravsray.github.io/MA-5105-Aug-26/”
- Codes corresponding to each lecture will be available at the same place.

Prerequisites and Demands

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- Some python programming experience
- Basic linear algebra and calculus

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Course Policies

Collaboration

- struggle first

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- discuss specific doubts over the macroscopic problem

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Assignments

- No assignments

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Assignments

- No assignments
- A lot of problem sets

Grading

- 4 surprise quizzes (fairly easy), 10 Marks each.
- 1 viva exam, 20 Marks.
- Class participation: 20 Marks.
- A final project worth 20 Marks.

Computing

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  2. logical : Compare / And / Or
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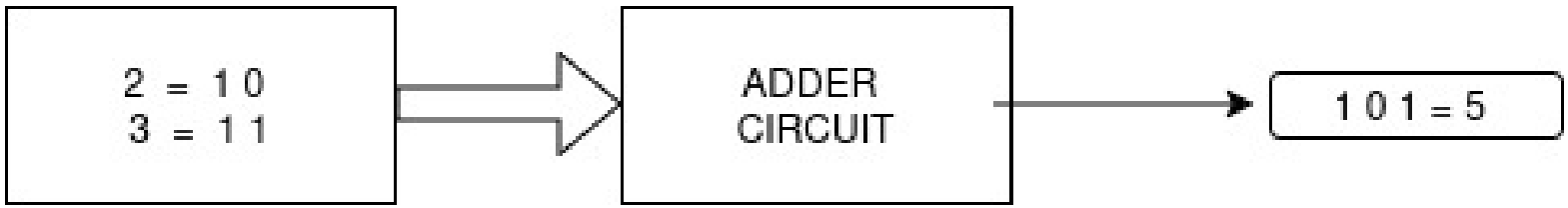

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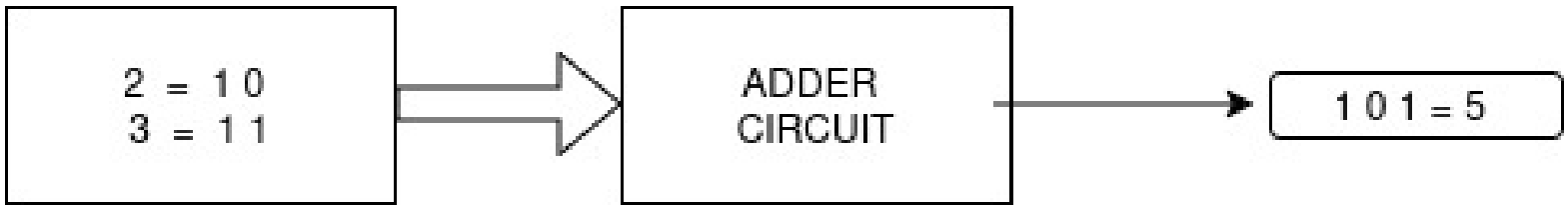

- By itself not very impressive.

Computing

Example-2:

1. Read marks of next student;
2. Add score;
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- programming language provides a set of primitive operations
- create ”expressions” which are complex but legal combination of primitives
- write down a sequence of declaration / evaluation usually to meet some end goal.

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- programming language provides a set of primitive operations
- create ”expressions” which are complex but legal combination of primitives
- write down a sequence of declaration / evaluation usually to meet some end goal.
- Complex software is built by combining numerous of these tiny, simple instructions into a carefully organised sequence.

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Scripts/Shells

- can be typed directly in a shell
- typically stored in a file

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- `sudo apt-get install idle-python3.12 python3-pip`
- `idle-python3.12` (start idle)

Where things may go wrong

Syntactic errors:

- common
- easily caught by compilers
- English:

  “sat cat mat” $\rightarrow$ not syntactically valid

  “cat sat on the mat” $\rightarrow$ syntactically valid
- Programming Language:

  “hi” 5 $\rightarrow$ not syntactically valid

  3.2*5 $\rightarrow$ syntactically valid

Where things may go wrong

Static semantic errors:

- syntactically valid strings
- no semantic errors
  1. program crashes, stops running
  2. program runs forever
  3. program gives an answer but different than expected

Content

Objects

Branching and Iterations

OBJECTS

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- objects have a type
  - Saurav is a human so he can breathe, sleep, etc.
  - An uncle is a relative, so he can give career advice, discuss politics, ask about your grades.
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- use `type (..)` to see the type of an object

Scalar objects

Type conversion (cast)

- can convert object of one type to another
- `float (3)` converts integer `3` to float `3.0`
- `int (3.9)` truncates float `3.9` to integer `3`

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Print to console

- `3 + 2`

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- `roi = 9.8`
- `prin = 1000.0`
- `time = 6`
- `interest = prin * roi * time / 100.0`
- can modify interest by modifying any of the basic components.

Math Functions and Number Methods

1. `round()`, for rounding numbers to some number of decimal places
   - ```
     >>> round(2.3)
     >>> round(3.14159, 3)
     >>> round(2.65, 1.4)
     ```
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   - `>>> round(2.3)`
     `>>> round(3.14159, 3)`
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2. `abs()`, for getting the absolute value of a number
   - `>>> abs(3)`
     `>>> abs(-5.0)`
3. `pow()`, for raising a number to some power
   - `>>> pow(2, 3)`
     `>>> pow(2, -2)`
     `>>> pow(2, 3, 2)`

Strings

- Many programmers deal with text on a daily basis
- example
  - web developers work with text that gets input from web forms
  - Data scientists process text to extract data and perform things like sentiment analysis, to classify opinions in a body of text.

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  - web developers work with text that gets input from web forms
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- Collections of text in Python are called strings

Strings

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  `hi = 'hello there'` or

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```
hi = 'hello there' or
hi = "hello there" or
```

- concatenate strings

```
name = "saurav"
greet = hi + name
greeting = hi + " " + name
```

- String Indexing

```
>>> flavor = "apple pie"
>>> flavor[1]
'p'
```

- In Python—and most other programming languages—counting always starts at zero.

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- If we want to alter a string, we must create an entirely new string

Converting Strings to Numbers

- `int()` stands for integer and converts objects into whole numbers,
- `float()` stands for floating-point number and converts objects into numbers with decimal points.

Strings

Converting Numbers to Strings

- Sometimes you need to convert a number to a string.
- ```
  >>> num_pancakes = 10
  >>> "I am going to eat " + num_pancakes
  + " pancakes."
  ```
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  ```
- Since `num_pancakes` is a number, Python can't concatenate it with the string "I'm going to eat".
- To build the string, you need to convert `num_pancakes` to a string using `str()`:

  ```
  >>> num_pancakes = 10
  >>> "I am going to eat " +
  str(num_pancakes) + " pancakes."
  ```
  ```

Input / Output

- output

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>>> x = 1
>>> print(3)
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>>> x = 1
>>> print(3) >>> print(5*text)
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Interact With User Input

- input

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>>> text = input("Type anything...  ")
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- Working With Strings and Number:
  user input using the `input()` function, the result is always a string.

Content

Objects

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Comments

```
import math

principal = 500000
interest = 7.5
duration = 20

r = interest / 12 / 100
n = duration * 12

if r == 0:
    payment = principal / n
else:
    payment = principal * r * (1 + r) ** n / ((1 + r) ** n - 1)

print("Monthly EMI = ", payment, " Ruppes")
```

Comments

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A person can understand the syntax, but they may wonder:

- Why is the interest divided by 12 and then by 100?
- Where did this complicated formula come from?
- Why is there a special case when r == 0?

The code works, but the intent is not obvious.

Comments

```python
import math

# Loan details
principal = 500000        # Loan amount in rupees
interest = 7.5            # Annual interest rate (in percent)
duration = 20             # Loan repayment period (in years)

# Convert the annual interest rate into a monthly decimal rate,
# as the EMI formula works with monthly interest.
r = interest / 12 / 100

# Total number of monthly installments.
n = duration * 12

# If there is no interest, each installment is simply an equal
# share of the principal.
if r == 0:
    payment = principal / n
else:
    # Compute the Equated Monthly Installment (EMI) using the
    # standard loan amortization formula used by banks.
    payment = principal * r * (1 + r) ** n / ((1 + r) ** n - 1)

# Display the monthly payment amount.
print("Monthly EMI =", payment, "Rupees")
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What the comments add

- Why the interest rate is converted.
- What n represents.
- Why there is a special case for zero interest.
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- Why the interest rate is converted.
- What n represents.
- Why there is a special case for zero interest.
- Where the formula comes from (the standard EMI formula).
- What each input variable means.

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- Use comments only when they add value

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- Functions break code into smaller chunks, and are great for tasks that a program uses repeatedly

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- Now `len` has a string value
- The variable name `len` is a keyword in Python, and even though you can change it's value, it's usually a bad idea to do so.

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- The del keyword is used to un-assign a variable from a value.
- `del` stands for delete, but it doesn't delete the value. Instead, it detaches the name from the value and deletes the name.

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- When a function is done executing, it returns a value as output.

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2. The function executes, and some action is performed with the arguments.
3. The function returns, and the original function call is replaced with the return value.

```
>>> num_letters = len("four")
```

- When a function changes or affects something external to the function itself, it is said to have a side effect.

  ```
  >>> return_value= print("What do I return?")
  ```

User define Functions

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- Repetitive code can be a nightmare to maintain.
- If you find a mistake in some code that's been copied and pasted all over the place, you'll end up having to apply the fix everywhere the code was copied.

The Anatomy of a Function

Every function has two parts:

1. The **function signature** defines the name of the function and any inputs it expects.

User define Functions

- Repetitive code can be a nightmare to maintain.
- If you find a mistake in some code that's been copied and pasted all over the place, you'll end up having to apply the fix everywhere the code was copied.

The Anatomy of a Function

Every function has two parts:

1. The **function signature** defines the name of the function and any inputs it expects.
2. The **function body** contains the code that runs every time the function is used.

Multiplication function

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- The first line of code in a function is called the function signature.
- It always starts with the `def` keyword, which is short for “define.”

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- The `def` keyword
- The function name, `multiply`
- The parameter list, `(x, y)`
- A colon `(:)` at the end of the line

CONTROL FLOW - BRANCHING

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if age >= 18:
    print("Eligible to vote")

print("Program finished")
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- In Python, indentation is mandatory, not just a matter of style.

Indentation in Python

Incorrect Indentation

```
marks = 45

if marks >= 50:
     print("Marks = ", marks)
print("Pass")
```